

# Ballistic Exercise as a Pre-Activation Stimulus: A Review of the Literature and Practical Applications

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**Abstract** Post-activation potentiation (PAP) refers to the acute enhancement of muscular function as a direct result of its contractile history. Protocols designed to elicit PAP have commonly employed heavy resistance exercise (HRE) as the pre-activation stimulus; however, a growing body of research suggests that low-load ballistic exercises (BE) may also provide an effective stimulus. The ability to elicit PAP without the need for heavy equipment would make it easier to utilise prior to competition. It is hypothesised that BE can induce PAP given the high recruitment of type II muscle fibres associated with its performance. The literature has reported augmentations in power performance typically ranging from 2 to 5 %. The performance effects of BE are modulated by loading, recovery and physical characteristics. Jumps performed with an additional loading, such as depth jumps or weighted jumps, appear to be the most effective activities for inducing PAP. Whilst the impact of recovery duration on subsequent performance requires further research, durations of 1–6 min have been prescribed successfully in multiple instances. The effect of strength and sex on the PAP response to BE is not yet clear. Direct comparisons of BE and HRE, to date, suggest a tendency for HRE protocols to be more effective; future research should consider that these strategies must be optimised in different ways. The role of acute augmentations in lower limb stiffness is

proposed as an additional mechanism that may further explain the PAP response following BE. In summary, BE demonstrates the potential to enhance performance in power tasks such as jumps and sprints. This review provides the reader with some practical recommendations for the application of BE as a pre-activation stimulus.

## Key Points

Post-activation potentiation (PAP) acutely enhances short-duration athletic performances that require maximal power production.

Ballistic exercise-based PAP-induced improvements in performance range from 2 to 5 % and are not dissimilar to those induced by heavy resistance exercise.

Ballistic exercise protocols that employ either depth jumps or weighted jumps (including weightlifting variations) appear to be the most effective.

## 1 Introduction

Post-activation potentiation (PAP) refers to the acute enhancement of muscular function as a direct result of its contractile history, for example, an augmentation of power output following a pre-conditioning contraction [1, 2]. Heavy resistance exercise (HRE) involves the performance of a multi-joint free weight exercise at loads typically exceeding 85 % 1 repetition maximum (1RM). PAP has commonly been utilised through the medium of complex training, where HRE is performed prior to a matched,

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higher velocity power exercise, for instance, a back squat followed by a countermovement jump (CMJ), with a view to augmenting performance in the second activity [2]. However, the practicality of employing HRE prior to performance may be questioned. These pre-activation protocols require heavy loadings, commonly approaching twice an athlete's bodyweight (e.g. Kilduff et al. [3]), and cumbersome equipment such as squat stands. Moreover, athletes competing in certain sports, such as athletics, will be required to report to a holding area before their event, which would typically render the performance of such activities impossible.

Ballistic exercise (BE) is characterised by the intention to perform a movement with maximal velocity [4] and by the acceleration of a mass throughout an entire movement [5]. BE may provide a viable alternative to HRE if it can be implemented correctly; such activities also seek to achieve maximal motor unit recruitment, as described in Sect. 2 of this review, but may be performed without the necessity for heavy and cumbersome equipment. This review will critically evaluate the potential for BE to augment subsequent performances and highlight their potential application within sports training and competition.

### 1.1 Literature Search Methodology

Original journal articles were retrieved from electronic searches of Science Direct, OVIDSP, Medline (EBSCO) and PubMed databases in addition to searches of Google Scholar and relevant bibliographic hand searches with no limits of language of publication. The search strategy included the terms post-activation potentiation, PAP, pre-conditioning, pre-activation, potentiating, dynamic warm-up, loaded warm-up, weighted jump, and loaded jump. The month of the last search performed was August 2013. Articles that investigated the effect of BE on performance were considered eligible for this review; inclusion criteria stated that articles must use quantifiable performance measures as a dependent variable.

## 2 Theoretical Basis of Ballistic Exercise (BE) Pre-Activation Strategies

The background and underlying mechanisms of PAP have been discussed in a number of review articles, including those by Hodgson et al. [2] and Tillin and Bishop [6]. These articles consider the potential for contribution from the phosphorylation of myosin regulatory light chains, increases in fascicle pennation angle and the recruitment of higher order motor units; readers are directed to these texts for more detailed consideration of these mechanisms as here we focus on its practical application.

It is important to note that whilst potentiation of muscle twitch is greatest immediately following the pre-activation stimulus [7–10], the same cannot be said for the performance benefit. The pre-activation stimulus will ultimately produce a certain level of fatigue alongside any potential PAP effect; whether or not the activity has a beneficial performance effect is governed by an interaction between these two responses. The role of recovery will be discussed in Sect. 3.2 of this review.

### 2.1 Type II Muscle Fibres

It may be hypothesised that the degree of muscular recruitment achieved by HRE is of key importance in determining its potential for PAP; research demonstrates that heavier loadings induce more favourable adaptations than lighter loadings [11–15]. Henneman's size principle [16, 17] suggests that a heavier loading will result in superior activation of the motor units comprising type II muscle fibres than a lighter loading. In vitro, these type II muscle fibres have been shown to carry a greater potential for PAP [18–20] and this explains why heavier loadings should theoretically induce a greater PAP response.

### 2.2 Ballistic Contraction

A ballistic contraction may be defined by the intention to perform movement with maximal velocity [4]. An acknowledged pre-requisite of ballistic exercise is that a mass is accelerated throughout the entire movement [5]; BE must therefore involve either a jumping action where the body leaves the floor or a throwing action where the projectile leaves the hand. BE removes the braking phase associated with traditional resistance exercise and has been suggested by Newton et al. [5] to increase the relative duration of positive acceleration and thereby facilitate greater force output and muscle activation [5]. Whilst these assertions have been contested by the likes of Frost et al. [21] and Lake et al. [22], BE still appears to facilitate greater power outputs than the same exercise performed in a non-ballistic manner [22].

During ballistic contraction, the threshold of motor unit recruitment is lower than in slower, ramped contraction [4, 23, 24]. It may be postulated that this reduction in recruitment threshold is the primary reason why BE may provide an effective stimulus for PAP; the strong excitatory drive of ballistic contraction enables the entire motor-neurone pool to be activated within a few milliseconds [25]. Whilst the recruitment threshold during ballistic contraction is lower than in ramped contraction [4, 23, 24], there does not appear to be a selective recruitment of faster motor units and the size principle of contraction is largely preserved [4, 23, 24, 26].

## 2.3 Overview of BE as a Pre-Activation Protocol

A compilation of the published, peer-reviewed research studies investigating the use of BE pre-activation protocols is shown in Table 1. The key modulating variables from these studies will be considered in Sect. 3 of this review.

## 3 Modulating Factors

### 3.1 Loading

As discussed, HRE protocols employing heavier loadings appear to induce more favourable acute adaptations than lower intensity [11–15]. Loading is also an important consideration when utilising BE, be this in relation to the choice of exercise used (i.e. plyometric intensity) or through a form of external loading such as a weighted jump or Olympic weightlifting variation.

#### 3.1.1 Plyometric Intensity

Read et al. [27] reported the performance of three maximal CMJs, recognised as a ‘moderate-intensity’ exercise [28], as inducing a 2.2 % ( $P < 0.05$ ; effect size [ES] 0.16) improvement in club head speed during a golf swing. Depth jumps are recognised as a ‘higher-intensity’ exercise [28] and may be predicted to carry a greater potential for PAP. For example, Masamoto et al. [29] demonstrated the performance of depth jumps to elicit a 4.9 kg (3.5 %;  $P < 0.05$ ; ES 0.16) increase in 1RM back squat, but a matched volume of tuck jumps (classified as a ‘lower-intensity’ exercise than depth jumps by Potach and Chu [28]) did not. Till and Cooke [30] and Tsolakis et al. [31] have also reported tuck jump protocols to carry no effect on performance. This is perhaps surprising given that a tuck jump should infer greater forces on landing due to the probable increase in height attained. It may therefore be that it is the intention to pre-activate and minimise the amortisation phase during a depth jump that provides a greater stimulus for PAP. Studies such as those conducted by Till and Cooke [30] and Tsolakis et al. [31] tend to instruct tuck jumps to be performed with a sole emphasis on achieving maximal jump height. This could imply a greater emphasis on the augmentation of limb stiffness, a concept discussed in Sect. 5 of this review, as a potential mechanism of PAP, although this would need to be examined directly before conclusions could be drawn.

Using depth jump protocols, Hilfiker et al. [32] observed a 2.2 % ( $P < 0.05$ ; ES 0.27) increase in average power production during a CMJ, Terzis et al. [33] a 4.6 % ( $P < 0.01$ ; ES 0.32) improvement in shot throw distance and Chen et al. [34] an improvement in vertical jump

height of  $\sim 1$  % (estimated from figures as means not provided;  $P = 0.008$ ). The greatest improvements have been observed by Lima et al. [35], presenting peak improvements of 6 % ( $P < 0.01$ ; ES 3.16) in CMJ and 2.7 % ( $P < 0.05$ ; ES 0.69) in 50 m sprint performance. Depth jumps were also used as the culmination of the plyometric protocol employed by Tobin and Delahunt [36]; the investigators reported peak improvements in jump performance of 4.8 % ( $P < 0.001$ ; ES 0.39).

Tillin and Bishop [6] propose that the increased eccentric pre-loading component associated with depth jumps, in comparison with normal CMJ protocols, may facilitate greater neural excitation, a conjecture supported by the findings of Masamoto et al. [29]. However, the traditional view that depth jumps are a higher-intensity exercise than CMJs (i.e. Potach and Chu [28]) may be contested. The intensity of plyometric exercise may be determined by mechanical outputs [37, 38] or muscle activation [39] and does not necessarily support this assumption [37–39]. For example, a tuck jump is associated with greater knee reaction force [38], quadriceps and gastrocnemius activation [39] and plantar flexion torque, impulse and power [37] than a depth jump. However, depth jumps may involve greater hip extension torque and power [37].

Future research should seek to compare different types of plyometric exercise and look to evaluate the different kinetic variables that may be subsequently affected. Given how individual exercises often display completely different kinetic profiles from one another [37–39], it is possible that different potentiative mechanisms may explain the performance enhancements observed.

#### 3.1.2 Weighted Jumps

The intensity of BE may be amplified by increasing system mass through the utilisation of an external loading. McBride et al. [40] were the first investigators to employ a weighted jump protocol; the authors had athletes perform jump squats in a Smith machine with loadings equal to 30, 55 and 80 % of their 1RM back squat but observed that this did not influence subsequent vertical jump or sprint performance. Using a weighted vest, a far more portable and easy to employ form of external loading, Faigenbaum et al. [41] compared dynamic warm-ups performed with loadings equivalent to 2 and 6 % body mass. Neither protocol augmented performance in a 10 yard sprint or in vertical and horizontal jump tests beyond an unloaded warm-up, although the investigators’ figures suggest a tendency for improved horizontal jump performance following the 2 % warm-up. Other investigators have since reported that higher weight vest loadings may elicit greater performance increases when worn only for a few specific warm-up exercises such as jumps and power skips [42, 43].

**Table 1** Overview of the studies investigating the effect of ballistic pre-activation strategies on performance

| References             | Participant characteristics (mean $\pm$ SD)                       | Pre-activation stimulus                    | Volume                | Recovery before testing | Performance test    | Impact on performance                                                                                              | Modulating factors    |
|------------------------|-------------------------------------------------------------------|--------------------------------------------|-----------------------|-------------------------|---------------------|--------------------------------------------------------------------------------------------------------------------|-----------------------|
| Masamoto et al. [29]   | 12 AT males                                                       | DJ                                         | 2 reps $\times$ 1 set | 30 s                    | 1RM BS              | 3.5 % (4.9 kg) increase from pre-test ( $P < 0.05$ ; ES 0.16; 90 % CI 0.9–8.9 kg)                                  | NR                    |
|                        | A: 20.5 $\pm$ 1.4 years                                           |                                            |                       |                         |                     |                                                                                                                    |                       |
|                        | H: 1.79 $\pm$ 0.11 m                                              | Tuck jump                                  | 3 reps $\times$ 1 set | 30 s                    | 1RM BS              | No effect                                                                                                          |                       |
|                        | BM: 87.4 $\pm$ 11.6 kg<br>CMJ: 0.60 $\pm$ 0.14 m                  |                                            |                       |                         |                     |                                                                                                                    |                       |
| Burkett et al. [54]    | 29 AT males                                                       | DB loaded box jump w/ 10 % BW (0.64 m box) | 5 reps $\times$ 1 set | 2 min                   | CMJ height          | 3.3 % (2.28 cm) increase over control ( $P < 0.01$ ; ES 0.38; 90 % CI 0.9–3.7 cm)                                  | NR                    |
|                        | A: 20.0 $\pm$ 1.7 years                                           |                                            |                       |                         |                     |                                                                                                                    |                       |
|                        | H: 1.90 $\pm$ 0.09 m<br>BM: 95.9 $\pm$ 13.0 kg                    | CMJ @ 75 % intensity                       | 5 reps $\times$ 1 set | 2 min                   | CMJ height          | No effect                                                                                                          |                       |
| McBride et al. [40]    | 15 AT males                                                       | BS w/ 90 % 1RM                             | 3 reps $\times$ 1 set | 4 min                   | 40 m sprint         | 0.87 % (0.05 s) faster than control ( $P = 0.018$ ; ES 0.16; 90 % CI 0.02–0.08 s)                                  | No effect of strength |
|                        | A: 20.8 $\pm$ 1.0 years                                           |                                            |                       |                         |                     |                                                                                                                    |                       |
|                        | H: 1.84 $\pm$ 0.07 m<br>M: 100.1 $\pm$ 15.5 kg<br>1RM BS/BM: 1.84 | Smith machine CMJ w/ 30 % of 1RM BS        | 3 reps $\times$ 1 set | 4 min                   | 40 m sprint         | No effect                                                                                                          |                       |
| Faigenbaum et al. [41] | 18 AT females                                                     | WV loaded DWU w/ 2 % BW                    | NR                    | 2 min                   | CMJ height          | No effect                                                                                                          | NR                    |
|                        | A: 15.3 $\pm$ 1.2 years                                           |                                            |                       |                         | LJ distance         | ~7 % increase over control (estimated from figures; $P$ value NR)                                                  |                       |
|                        | H: 1.66 $\pm$ 0.09 m                                              |                                            |                       |                         | MB throw            | No effect                                                                                                          |                       |
|                        | BM: 61.6 $\pm$ 10.4 kg                                            |                                            |                       |                         | 10 yard sprint      | No effect                                                                                                          |                       |
|                        |                                                                   | WV loaded DWU w/ 6 % BW                    | NR                    | 2 min                   | CMJ height          | No effect                                                                                                          |                       |
|                        |                                                                   |                                            |                       |                         | LJ distance         | No effect                                                                                                          |                       |
|                        |                                                                   |                                            |                       |                         | Medicine ball throw | No effect                                                                                                          |                       |
|                        |                                                                   |                                            |                       |                         | 10 yard sprint      | No effect                                                                                                          |                       |
| Hilfiker et al. [32]   | 13 AT males                                                       | Modified DJ from 0.6 m                     | 5 reps $\times$ 1 set | 1 min                   | CMJ height          | No effect                                                                                                          | NR                    |
|                        | A: 22 years (range 20–28)                                         |                                            |                       |                         | CMJ power           | 2.2% (1.21 W kg <sup>-1</sup> ) increase over control ( $P < 0.05$ ; ES 0.27; 90 % CI 0.2–2.2 W kg <sup>-1</sup> ) |                       |
|                        | BM: 69.5 kg (range 54.5–85)                                       |                                            |                       |                         | SJ height           | No effect                                                                                                          |                       |
|                        |                                                                   |                                            |                       |                         | SJ power            | No effect                                                                                                          |                       |
| Thompsons et al. [42]  | 16 AT female                                                      | DWU w/ 10 % BW                             | NR                    | 2 min                   | CMJ height          | No effect                                                                                                          | NR                    |
|                        | A: 19.7 $\pm$ 1.4 years                                           | WV for last 4 exercises                    |                       |                         | LJ distance         | 2.5 % (4.6 cm) increase over control ( $P \leq 0.05$ ; ES 0.24; 90 % CI 0.8–8.4 cm)                                |                       |
|                        | H: 1.66 $\pm$ 0.11 m<br>BM: 67.0 $\pm$ 10.7 kg                    |                                            |                       |                         |                     |                                                                                                                    |                       |

Table 1 continued

| References           | Participant characteristics (mean $\pm$ SD)                                                                                                                                        | Pre-activation stimulus                                                                                                                                                                                                                         | Volume                                                                  | Recovery before testing                  | Performance test                                                                    | Impact on performance                                                                                                                                                                                               | Modulating factors                                                                                                     |
|----------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------|------------------------------------------|-------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------|
| Tahayori [43]        | 10 PA males, 8 PA females<br>A: 21.7 $\pm$ 1.2 years<br>H: 1.67 $\pm$ 0.08 m<br>BM: 64.4 $\pm$ 16.3 kg                                                                             | WV loaded CMJ w/ 15 % BW                                                                                                                                                                                                                        | 3 reps $\times$ 5 sets (0.5 min recovery)                               | 2 min                                    | CMJ height                                                                          | 2.1 % (1.48 cm) increase over control ( $P \leq 0.05$ ; SD NR; 90 % CI 0.3–2.7 cm) in males<br>No effect in females                                                                                                 | Effect of gender                                                                                                       |
| Terzis et al. [33]   | 8 PA males<br>A: 22.0 $\pm$ 1.0 years<br>H: 1.77 $\pm$ 0.05 m<br>BM: 77.0 $\pm$ 6.0 kg<br>8 PA females<br>A: 23.0 $\pm$ 3.0 years<br>H: 1.70 $\pm$ 0.06 m<br>BM: 66.0 $\pm$ 7.0 kg | DJ from 0.4 m                                                                                                                                                                                                                                   | 5 reps $\times$ 1 set                                                   | 20 s                                     | Underhand front shot distance                                                       | 4.6 % (0.38 m) increase from pre-test ( $P < 0.01$ ; ES 0.32; 90 % CI 0.2–0.6 m)<br>[1.5 % (0.11 m) increase in females (ns; ES 0.14); 7.4 % (0.64 m) increase in males ( $P < 0.01$ ; ES 0.69; 90 % CI 0.3–1.0 m)] | Effect of gender<br>Effect of strength ( $r = 0.50$ ; $P < 0.05$ )<br>Effect of fibre type ( $r = 0.76$ ; $P < 0.01$ ) |
| Till and Cooke [30]  | 12 AT males<br>A: 18.3 $\pm$ 0.72 years<br>H: 1.76 $\pm$ 0.05 m<br>BM: 72.1 $\pm$ 8.0 kg                                                                                           | Deadlift w/ 5RM<br>Tuck jump<br>Isometric MVC (3 s)                                                                                                                                                                                             | 5 reps $\times$ 1 set<br>5 reps $\times$ 1 set<br>3 reps $\times$ 1 set | 4–6 min<br>7–9 min<br>4–6 min<br>7–9 min | 20 m sprint<br>CMJ height<br>20 m sprint<br>CMJ height<br>20 m sprint<br>CMJ height | No effect<br>No effect<br>No effect<br>No effect<br>No effect<br>No effect                                                                                                                                          | No effect of strength                                                                                                  |
| Chattong et al. [44] | 20 AT males<br>A: 22.45 $\pm$ 1.73 years<br>H: 1.77 $\pm$ 0.07 m<br>BM: 76.98 $\pm$ 8.56 kg                                                                                        | Box jump (box @ knee height)<br>WV loaded box jump w/ 5 % BW (box @ knee height)<br>WV loaded box jump w/ 10 % BW (box @ knee height)<br>WV loaded box jump w/ 15 % BW (box @ knee height)<br>WV loaded box jump w/ 20 % BW (box @ knee height) | 5 reps $\times$ 1 set                                                   | 2 min                                    | CMJ height                                                                          | No effect<br>No effect<br>No effect<br>No effect<br>No effect                                                                                                                                                       | NR                                                                                                                     |

**Table 1** continued

| References               | Participant characteristics (mean $\pm$ SD) | Pre-activation stimulus | Volume                 | Recovery before testing | Performance test | Impact on performance                                                            | Modulating factors  |
|--------------------------|---------------------------------------------|-------------------------|------------------------|-------------------------|------------------|----------------------------------------------------------------------------------|---------------------|
| McCann and Flanagan [47] | 8 AT males                                  | Hang clean w/ 5RM       | 5 reps $\times$ 1 set  | 4 min                   | CMJ height       | No effect                                                                        | No effect of gender |
|                          | A: 20.86 $\pm$ 1.77 years                   | Hang clean w/ 5RM       | 5 reps $\times$ 1 set  | 5 min                   |                  | No effect                                                                        |                     |
|                          | H: 1.95 $\pm$ 0.05 m                        | BS w/ 5RM               | 5 reps $\times$ 1 set  | 4 min                   |                  | No effect                                                                        |                     |
|                          | BM: 89.42 $\pm$ 8.40 kg                     | BS w/ 5RM               | 5 reps $\times$ 1 set  | 5 min                   |                  | No effect                                                                        |                     |
|                          | 8 AT females                                |                         |                        |                         |                  |                                                                                  |                     |
|                          | A: 19.14 $\pm$ 0.38 years                   |                         |                        |                         |                  |                                                                                  |                     |
| Andrews et al. [46]      | H: 1.77 $\pm$ 0.07 m                        |                         |                        |                         |                  |                                                                                  | NR                  |
|                          | BM: 72.27 $\pm$ 9.41 kg                     |                         |                        |                         |                  |                                                                                  |                     |
|                          | 19 AT females                               | Hang clean w/ 60 % 1RM  | 3 reps $\times$ 3 sets | 3 min after set 1       | CMJ height       | No effect                                                                        |                     |
|                          | A: 20.5 $\pm$ 1.5 years                     |                         |                        | 3 min after set 2       |                  | No effect                                                                        |                     |
|                          | H: 1.72 $\pm$ 0.06 m                        |                         |                        | 3 min after set 3       |                  | No effect                                                                        |                     |
|                          | BM: 70.2 $\pm$ 8.3 kg                       |                         |                        |                         |                  |                                                                                  |                     |
|                          |                                             | BS w/ 75 % 1RM          | 3 reps $\times$ 3 sets | 3 min after set 1       | CMJ height       | No effect                                                                        |                     |
|                          |                                             |                         |                        | 3 min after set 2       |                  | No effect                                                                        |                     |
|                          |                                             |                         |                        | 3 min after set 3       |                  | No effect                                                                        |                     |
| Lima et al. [35]         | 10 AT males                                 | DJ from 0.75 m          | 5 reps $\times$ 2 sets | 5 min                   | CMJ height       | No effect                                                                        | NR                  |
|                          | A: 20.6 $\pm$ 2.6 years                     |                         |                        |                         | 50 m sprint      | No effect                                                                        |                     |
|                          | H: 1.76 $\pm$ 0.06 m                        |                         |                        | 10 min                  | CMJ height       | No effect                                                                        |                     |
|                          | BM: 73.7 $\pm$ 9.2 kg                       |                         |                        |                         | 50 m sprint      | 2.4 % (0.16 s) faster than control ( $P < 0.05$ ; ES 0.66; 90 % CI 0.03–0.3 s)   |                     |
|                          |                                             |                         |                        | 15 min                  | CMJ height       | 5.5 % (2.4 cm) increase over control ( $P < 0.01$ ; ES 3.16; 90 % CI 1.0–3.8 cm) |                     |
|                          |                                             |                         |                        |                         | 50 m sprint      | 2.7 % (0.17 s) faster than control ( $P < 0.05$ ; ES 0.69; 90 % CI 0.03–0.3 s)   |                     |

Table 1 continued

| References             | Participant characteristics (mean $\pm$ SD)                                                                                       | Pre-activation stimulus                                             | Volume                                  | Recovery before testing | Performance test                                       | Impact on performance                                                                    | Modulating factors                                                                   |
|------------------------|-----------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------|-----------------------------------------|-------------------------|--------------------------------------------------------|------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------|
| Tsolakakis et al. [31] | 13 AT males<br>A: 21.8 $\pm$ 3.7 years<br>H: 1.79 $\pm$ 0.05 m<br>BM: 76.1 $\pm$ 7.8 kg<br>1RM leg press: 219 $\pm$ 44 kg         | Isometric leg press at 90° knee flexion (3 s)                       | 3 reps $\times$ 1 set                   | 8 min                   | CMJ power                                              | No effect overall<br>[7.5 % decrease in males ( $P < 0.01$ ; ES 0.53)]                   | Effect of gender, negative effect of strength for males ( $r = -0.55$ ; $P < 0.05$ ) |
|                        |                                                                                                                                   |                                                                     |                                         | 12 min                  |                                                        | No effect overall<br>[8.7 % decrease in males ( $P < 0.01$ ; ES 0.65)]                   |                                                                                      |
|                        | 10 AT females<br>A: 22.7 $\pm$ 4.8 years<br>H: 1.69 $\pm$ 0.07 m<br>BM: 60.9 $\pm$ 4.6 kg<br>1RM leg press: 118 $\pm$ 29 kg       | Tuck jump                                                           | 5 reps $\times$ 3 sets (1 min recovery) | 8 min                   | CMJ power                                              | No effect                                                                                | No effect of gender                                                                  |
|                        |                                                                                                                                   |                                                                     |                                         | 12 min                  |                                                        | No effect                                                                                |                                                                                      |
| Chiu and Salem [45]    | 13 AT males<br>A: 27.31 $\pm$ 4.21 years<br>H: 1.79 $\pm$ 0.11 m<br>BM: 96.09 $\pm$ 17.61 kg<br>1RM snatch: 123.46 $\pm$ 19.99 kg | Snatch pull (1 set = 2 reps each @ 70, 80, 90, 100 % of 1RM snatch) | 8 reps $\times$ 2 sets                  | 3 min after set 1       | CMJ height                                             | 5.8 % increase from pre-test ( $P = 0.001$ ; ES 1.62; mean figures NR)                   | NR                                                                                   |
|                        |                                                                                                                                   |                                                                     |                                         | 3 min after set 2       |                                                        | 5.9 % increase from pre-test ( $P < 0.001$ ; ES 1.75; mean figures NR)                   |                                                                                      |
| West et al. [58]       | 20 AT males<br>A: 26.5 $\pm$ 4.1 years<br>H: 1.85 $\pm$ 0.09 m<br>BM: 97.3 $\pm$ 12.5 kg<br>1RM bench press: 144 $\pm$ 19 kg      | Bench press w/ 87 % 1RM                                             | 3 reps $\times$ 3 sets (recovery NR)    | 8 min                   | Ballistic bench throw power w/ 30 % of 1RM bench press | 4.3 % (38 W) increase from pre-test ( $P < 0.001$ ; ES 0.35; 90 % CI 21–55 W)            | Effect of strength ( $r = 0.68$ ; $P = 0.001$ )                                      |
|                        |                                                                                                                                   | Ballistic bench throw w/ 30 % of 1RM bench press                    | 3 reps $\times$ 3 sets (recovery NR)    | 8 min                   | Ballistic bench throw power w/ 30 % of 1RM bench press | 3.6 % (32 W) increase from pre-test ( $P < 0.001$ ; ES 0.28; 90 % CI 18–46 W)            | Effect of strength ( $r = 0.63$ ; $P = 0.003$ )                                      |
| Chen et al. [34]       | 10 AT males<br>A: 20.9 $\pm$ 1.6 years<br>H: 1.78 $\pm$ 0.07 m<br>BM: 73.6 $\pm$ 10.7 kg                                          | DJ from 0.6 m                                                       | 5 reps $\times$ 1 set                   | 2 min                   | CMJ height                                             | Increase from pre-test when combined w/ 2 set condition ( $P = 0.008$ ; mean figures NR) | NR                                                                                   |
|                        |                                                                                                                                   |                                                                     |                                         | 6 min                   |                                                        | No effect                                                                                |                                                                                      |
|                        |                                                                                                                                   | DJ from 0.6 m                                                       | 5 reps $\times$ 2 sets (1 min recovery) | 12 min                  |                                                        | No effect                                                                                |                                                                                      |
|                        |                                                                                                                                   |                                                                     |                                         | 2 min                   | CMJ height                                             | Increase from pre-test when combined w/ 2 set condition ( $P = 0.008$ ; mean figures NR) |                                                                                      |
|                        |                                                                                                                                   |                                                                     |                                         | 6 min                   |                                                        | No effect                                                                                |                                                                                      |
|                        |                                                                                                                                   |                                                                     |                                         | 12 min                  |                                                        | No effect                                                                                |                                                                                      |



Table 1 continued

| References              | Participant characteristics (mean $\pm$ SD)                                                                             | Pre-activation stimulus                                      | Volume                                                                     | Recovery before testing | Performance test    | Impact on performance                                                                                                                                                                                                                                             | Modulating factors |
|-------------------------|-------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------|----------------------------------------------------------------------------|-------------------------|---------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------|
| Read et al. [27]        | 16 PA males<br>A: $20.1 \pm 3.2$ years<br>H: $1.76 \pm 0.07$ m<br>BM: $72.8 \pm 7.8$<br>Handicap: $5.8 \pm 2.2$ strokes | CMJ                                                          | 3 reps $\times$ 1 set                                                      | 1 min                   | Golf clubhead speed | 2.2 % (2.25 mph) increase over control ( $P < 0.05$ ; ES 0.16; 90 % CI 0.4–4.1 mph)                                                                                                                                                                               | NR                 |
| Tobin and Delahunt [36] | 20 AT males<br>A: $22.4 \pm 3.4$ years<br>H: $1.84 \pm 0.07$ m<br>BM: $101.2 \pm 11.9$ kg                               | Ankle hops<br>Hurdle hops (hurdles @ 0.7 m)<br>DJ from 0.5 m | 10 reps $\times$ 2 sets<br>5 reps $\times$ 3 sets<br>5 reps $\times$ 1 set | 1 min<br>3 min<br>5 min | CMJ height          | 4.8 % (2.09 cm) increase from pre-test ( $P < 0.001$ ; ES 0.39; 90 % CI 1.2–3.0 cm)<br>3.9 % (1.72 cm) increase from pre-test ( $P < 0.001$ ; ES 0.31; 90 % CI 1.0–2.5 cm)<br>3.5 % (1.53 cm) increase from pre-test ( $P < 0.001$ ; ES 0.27; 90 % CI 0.9–2.2 cm) | NR                 |

A age, AT athletically trained, BM body mass, BS back squat, BW body weight, CI confidence interval, CMJ countermovement jump, DB dumbbell, DJ depth jump, DWU dynamic warm-up, ES effect size, H height, kg kilograms, LJ long jump, m metres, mph miles per hour, MVC maximum voluntary contraction, NR not reported, ns not significant, PA physically active, reps repetitions, SD standard deviation, SJ squat jump, W watts, w/ with, WV weighted vest,  $W\ kg^{-1}$  watts per kg of BM, IRM 1 repetition maximum

Thompsons et al. [42] demonstrated a 2.5 % ( $P \leq 0.05$ ; ES 0.24) improvement in horizontal, but not vertical, jump performance with a 10 % body mass loading and Tahayori [43] a 2.1 % ( $P \leq 0.05$ ; standard deviations not reported) improvement in vertical jump performance in males, but not females. Most recently, Chattong et al. [44] observed that loadings of 5–20 % had no impact on subsequent jump performance.

### 3.1.3 Olympic Weightlifting

Three studies have used Olympic weightlifting variations as a pre-activation stimulus [45–47]. Whilst Chiu and Salem [45] observed a 5.9 % ( $P \leq 0.01$ ; ES 1.75) improvement in jump height following the performance of heavy snatch pulls, such improvements were not echoed in the work of McCann and Flanagan [47] and Andrews et al. [46], who utilised power clean and hang clean exercises, respectively. Pull variations of Olympic weightlifting movements, such as those used by Chiu and Salem [45], are exercises that use the double knee bend and triple extension component of the full Olympic lifts but do not require the athlete to descend underneath and catch the bar [48]. Pull variations may be performed with higher loadings than the full Olympic lifts and facilitate the production of greater peak forces [49, 50]; it may therefore be hypothesised that they carry a greater potential for PAP. However, whether pull variations are indeed a more effective pre-activation stimulus than the full Olympic lifts would need to be established through direct comparison. As with HRE, Olympic weightlifting requires heavy weights and specialised equipment (i.e. a weightlifting platform). This means that they are also unlikely to be practical for use prior to competition by the majority of athletes. It would be interesting to investigate whether kettlebell exercises that mimic the ballistic triple extension of ankle, knee and hip associated with Olympic weightlifting movements may also be used as a pre-activation stimulus. For example, the two-handed kettlebell swing has been demonstrated to elicit impulses and power outputs comparable to those of weighted squat jumps [51] and may therefore provide an effective stimulus for PAP. Nonetheless, Olympic weightlifting movements may be suitably employed to elicit PAP during training.

### 3.1.4 Summary

In summary, it appears that increasing the loading of BE heightens the potential PAP response. However, as this may be associated with a concomitant increase in fatigue, which may mask the potential for performance enhancement, the trade-off between loading and the recovery time necessary to observe a beneficial performance effect must be considered.



### 3.2 Recovery

In studies that have directly examined the effect of recovery duration following HRE, a level of consensus has been reached. Performances are initially impaired by HRE, PAP is then realised, peaks and then decreases in an inverted U fashion [3, 11, 52, 53]. However, the same level of agreement cannot be reached in regards to BE with the reporting of equivocal results. The findings of Lima et al. [35] suggest that drop jumps may exhibit a similar recovery profile to HRE; they observed a tendency for vertical jump height and sprint performance to be impaired following 5 min of recovery before realising potentiation in jump performance after 15 min and in sprint performance after 10 and 15 min. Conversely, Chen et al. [34] demonstrated a negative association between recovery and PAP; vertical jump height 2 min after a drop jump protocol was greater than at 6 ( $P = 0.004$ ) and 12 min ( $P = 0.002$ ), jump height at 6 min was also greater than at 12 min ( $P = 0.018$ ); mean figures were not reported. The impact of recovery duration has also been evaluated by Tsolakis et al. [31] and Tobin and Delahunt [36]. Tsolakis et al. [31] reported no effect of upper- (ballistic push up) or lower-body (tuck jump) BE over a 12 min recovery period. Tobin and Delahunt [36] observed similar augmentations in vertical jump height at 1, 3 and 5 min following a protocol incorporating different types of vertical jumps.

It would appear that the recovery duration necessary to observe a PAP response may be lower than would be required following HRE given substantial reductions in system mass loading [11]. For example, several investigators have successfully employed durations of  $\leq 60$  s [27, 29, 32, 33, 36], 2 min [34, 42, 43, 54] and 3 min [36, 45]. Further research is certainly required to determine an optimal recovery period following a BE and should also attempt to investigate how this may be impacted by the intensity and volume of the protocol.

### 3.3 Physical Characteristics

The impact of individual differences in determining the PAP response has been widely investigated, although primarily in relation to HRE. It appears that athletes' strength levels are the most important consideration. Chiu et al. [55] reported that athletically trained males and females experienced a potentiation effect following HRE. However, the jump squat performance of recreationally trained males and females decreased following HRE. Chiu et al. [55] cite greater muscle activation in the athletic-trained population as the reason for this difference in response. Similar findings were noted by Gourgoulis et al. [56]. Moderate to strong correlations between strength and PAP responses following HRE have been reported in strength-trained

athletes ( $r = 0.49$ – $0.81$ ;  $P \leq 0.05$  [3, 52, 53, 57]). Terzis et al. [33] and West et al. [58] also report correlations of a similar magnitude following BE ( $r = 0.50$ ,  $P < 0.05$  and  $r = 0.626$ ;  $P = 0.003$ , respectively). However, when categorising their athletes into 'strong' and 'weak' subgroups, McBride et al. [40] (1RM squat/body mass:  $2.02 \pm 0.15$  vs.  $1.68 \pm 0.14$  kg) and Till and Cooke [30] (5RM deadlift/body mass:  $72.5 \pm 8.22$  vs.  $62.5 \pm 8.80$  kg) reported no effect of strength levels.

Given the role of type II muscle fibres in the PAP response [18–20], and the strong relationship established between strength and percentage of type II fibres [59], the apparent positive potentiation effect that HRE has for stronger athletes may be explained through fibre type percentage [6]. Indeed, Terzis et al. [33] saw stronger correlations for PAP and type II fibre percentage ( $r = 0.76$ ,  $P < 0.01$ ) than for strength. Furthermore, Jo et al. [60] have proposed that stronger individuals require less recovery to be able to benefit from PAP. Jo et al. [60] demonstrated the recovery duration (5, 10, 15 and 20 min durations were examined) eliciting the greatest improvements in a Wingate cycle test to be significantly correlated with 1RM back squat ( $r = -0.77$ ,  $P < 0.05$ ). It may be that higher strength does not directly influence recovery per se, but rather, indicates a higher training age and level of training tolerance.

Sex is another important consideration in regards to the modulation of PAP, although any such discrepancy may also be partially attributable to factors of strength or muscle fibre type. Rixon et al. [61] have demonstrated PAP to occur to significantly greater degrees in untrained or recreationally trained males than in matched-level females following HRE. Comyns et al. [62] report a similar tendency between resistance-trained male and female athletes. The findings of Tahayori [43] and Terzis et al. [33] suggest that similar inter-sex discrepancies may be prevalent following BE whilst those of McCann and Flanagan [47] do not.

Further investigation into the impact of subject characteristics on responses to BE is required before any definitive conclusions can be drawn. Given the apparent potential for limb stiffness to modulate PAP, a particular consideration of inter-subject differences in various components of limb stiffness is certainly warranted. The role of limb stiffness will be discussed later on in this review.

## 4 BE vs. Heavy Resistance Exercise (HRE) Protocols

HRE and plyometric protocols have been compared by McBride et al. [40] and Till and Cooke [30]. McBride et al. [40] observed an improvement in 40 m sprint time (0.87 %;  $P = 0.018$ ; ES 0.16) as a consequence of HRE,

but not BE. Till and Cooke [30] demonstrated that neither heavy deadlifts nor tuck jumps impacted sprinting and vertical jumping performance. Both sets of investigators suggested that their ballistic activities were not sufficient to elicit a PAP response in terms of volume and intensity, respectively, and proposed that their HRE protocols resulted in more motor unit recruitment. Again, the importance of physical characteristics can be noted, as well as the need to standardise this within research studies.

HRE and Olympic weightlifting protocols have been compared by McCann and Flanagan [47] and Andrews et al. [46]. Neither set of investigators detected PAP following variations of the clean exercise, although they did note differences between the two protocols. McCann and Flanagan [47] compared power cleans and heavy squats and reported an average improvement in jump height of 5.7 % ( $2.72 \pm 1.21$  cm;  $P < 0.001$ ; mean figures not reported) when subjects' preferred protocol was considered; for 5 of 14 subjects this was found to be the power clean. Andrews et al. [46] saw that jump performance over three sets was maintained following their hang clean protocol, but had decreased by set three following heavy squats. They conclude that Olympic weightlifting movements may therefore be better suited to complex training programmes than HRE.

As this review has highlighted, BE and HRE protocols must be optimised in different ways; for example, different intensities, volumes and recovery periods may be required. Future research should employ independently optimised BE and HRE protocols if seeking to directly compare these directly.

## 5 Limb Stiffness

Stiffness describes the resistance of an object to deformation [63] and changes in the centre of mass in response to force [64]. It is possible that acute augmentations in limb stiffness may serve as an additional mechanism in the explanation of PAP working alongside those that have been previously outlined. For example, HRE protocols have been shown to elicit acute increases in lower limb stiffness as determined by spring-mass models [13, 14]. During single leg drop jumps performed on a sledge apparatus Comyns et al. [13] detected a 10.9 % ( $P < 0.05$ ; ES 0.39) increase in leg-spring stiffness, which translated to a reduction in ground contact time of 7.8 % ( $P < 0.05$ ; ES 0.38), an indication of improved stretch-shortening cycle capabilities. Moir et al. [14] reported a 16 % ( $P = 0.013$ ; ES 0.52) increase in vertical stiffness during a vertical jump. In spite of these findings, any potential contribution for limb stiffness in the explanation of the PAP response has not been considered by previous review articles.

At the musculotendinous unit level, increases in stiffness would be expected to increase force development in the active component—as the muscle can function in a more quasi-isometric fashion—and increase the potential for elastic recoil from the passive component [64]. Increases in leg stiffness would therefore be expected to improve measures of reactive strength and the relative force contribution of the stretch-shortening cycle (i.e. contribution of passive tension to overall force production) during powerful movements given the viscoelastic properties of the musculotendinous unit [64, 65].

In spite of the increase in leg stiffness observed by Comyns et al. [13], the average flight time achieved during a single leg drop jump was reduced by 3.4 % ( $P < 0.01$ ; ES 0.58). Moir et al. [14] reported no effect of their HRE protocol on vertical jump height despite the observed increase in vertical stiffness. These findings highlight that acute increases in leg-spring stiffness may not directly correspond to an enhancement of performance. Arampatzis et al. [65] proposed that an inverted U relationship exists between leg-spring stiffness and mechanical power output during the propulsive phase of vertical drop jumping. Increases in leg-spring stiffness will enhance power output up until an athlete's optimal value is reached; further increases beyond this point will begin to impair power production. This decrease in power production is thought to be a likely consequence of an increase in muscle shortening velocity reducing the efficiency of force development [64]. Whilst this could explain why an augmentation in leg stiffness by Comyns et al. [13] and Moir et al. [14] did not transfer to an improvement in jump performance, it should be noted that both sets of investigators utilised relatively short recovery durations (4 and 2 min, respectively). Research suggests that longer recovery durations may be required following HRE in order to facilitate a benefit to performance [3, 11, 53].

Studies utilising BE protocols are yet to examine the impact on leg stiffness; however, as the muscle tendon unit will be required to stiffen to function effectively during these tasks [64], it may be possible that similar augmentations may be observed. This avenue could be explored by future research in order to be better understood.

## 6 Practical Application

### 6.1 Training

The application of PAP has been largely popularised through the application of complex training. The principle is that PAP will allow an athlete to train at power outputs exceeding their normal capabilities and therefore increase the potential training adaptations. The effectiveness of

**Table 2** An example of how a power-based training session could theoretically be structured. Exercises are repeated in a circuit format for 3–8 circuits

| Exercise type                 | Example exercise | Suggested prescription                       |
|-------------------------------|------------------|----------------------------------------------|
| Lower body ballistic exercise | Depth jump       | 3–5 repetitions<br>1–6 min post-set recovery |
| Lower body performance task   | 10 m sprint      | 1–3 repetitions<br>5–8 min post-set recovery |

complex training against other training modalities is yet to be properly determined; however, it would appear that a familiarisation period is necessary before the ergogenic potential of complex training may be realised [66, 67], and this should be reflected in the design of future studies. Although research has reached a general consensus that HRE can potentiate power-based activities, the potential for enhancing resistance exercise performance is an avenue largely unexplored by the research. The findings of Masamoto et al. [29] would suggest that this ‘reversal’ of complex pairs shows promise.

Table 2 shows an example of how a power-based training session could be theoretically structured to benefit from the PAP effect stimulated by BE. The suggested prescriptions are guided by the literature, but should be determined on an individual basis. As the recovery duration necessary to observe a PAP response following BE appears to be less than that following HRE, this type of session is a more time-efficient way of harnessing PAP than traditional complex training programmes. This type of protocol should therefore serve as a viable strategy for athletes and sports professionals to incorporate within their overall training programme. Also, because this programme is completed in a circuit format, it is interesting to ponder whether this protocol would be superior to traditional PAP protocols given the constant stimulatory rotation. Such suggestions warrant specific investigation if their validity is to be determined.

## 6.2 Competition

The performance enhancement effects of a traditional dynamic warm-up are well established (see Bishop [68] for a review); however, the application of protocols designed to elicit PAP may augment performances beyond those that may be achieved by warm-up alone [69]. Whilst the potential benefit of PAP to enhance short-duration, explosive activity (e.g. jumps and sprints) is clear, the impact this could have on intermittent activities (e.g. team sports) is much less so. Acute increases in motor potential may prove beneficial in almost all sports,

but the challenge, an almost impossible one at that, would be in maintaining a potentiated state for the duration of performance despite an inevitable accumulation of fatigue. It may be possible, and importantly worthwhile in some instances, to create a potentiated state for an athlete within which they can start their performance. This may give the athlete an initial advantage in competition and could indeed prove to be the difference between winning and losing.

## 7 Conclusion

PAP acutely enhances short-duration athletic performances that require maximal power production and may therefore benefit performance in sports where maximal power production is a key performance determinant. BE-based PAP-induced improvements in performance range from 2 to 5 % and are not dissimilar to those induced by HRE-based PAP-induced improvements in performance. BE protocols that employ either depth jumps or weighted jumps (including weightlifting variations) appear to be the most effective. Whilst the potential benefits of PAP to the individual athlete/s should be considered by the coach before seeking to apply this phenomenon, the performance of exercises such as depth jumps appear an easy-to-employ protocol with minimal logistical demand and minimal risk of performance detriment.

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